Statistics: Mini-Lessons

7.1 Scatter plots and correlation NC.8.SP.1

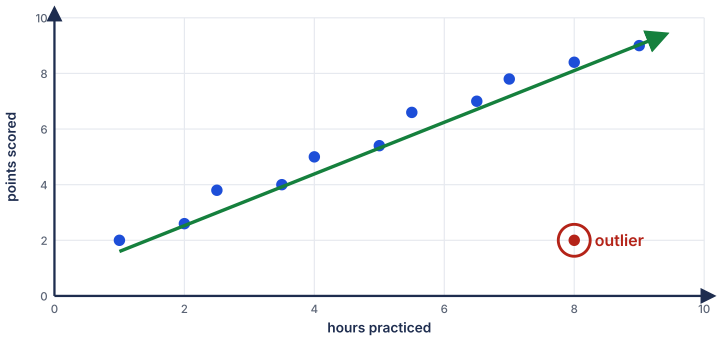
A **scatter plot** shows two sets of data as dots. Describe it: **direction** (positive = up, negative = down, none), **strength** (close to a line = strong), and **outliers** (dots far from the rest).

Positive correlation
as x goes up, y goes up

Negative correlation
as x goes up, y goes down

Outlier
a dot far away from the pattern

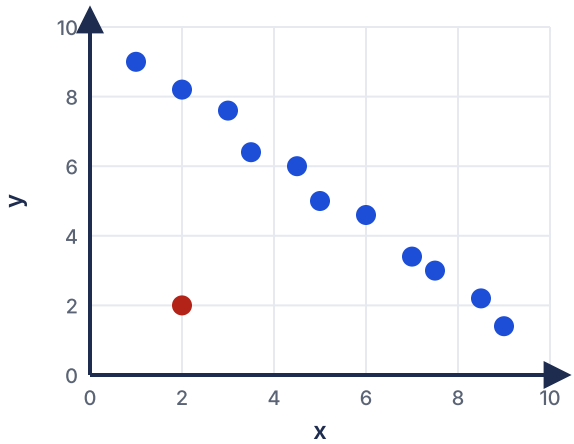
Correlation is not causation
two things moving together does not prove one causes the other



Left: negative (down). Right: no correlation (no pattern).

Watch out: Ice cream sales and sunburns both go up in summer. Ice cream does not **cause** sunburns: hot weather drives both.

Walk through: `Describe this scatter plot.`



1 Read left to right. Which way do most dots go?
Down: negative correlation → direction: negative

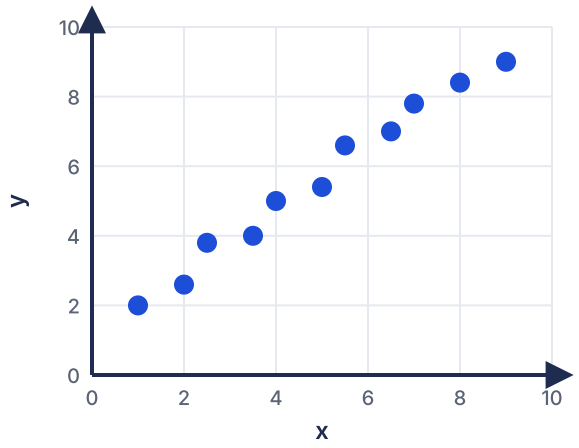
2 Are the dots close to a line or spread out?
Close to a line: strong → strength: strong

3 Is there an outlier?
Yes: the dot at (2, 2) → outlier: (2, 2)

Check your understanding

1 Score 3 NC.8.SP.1

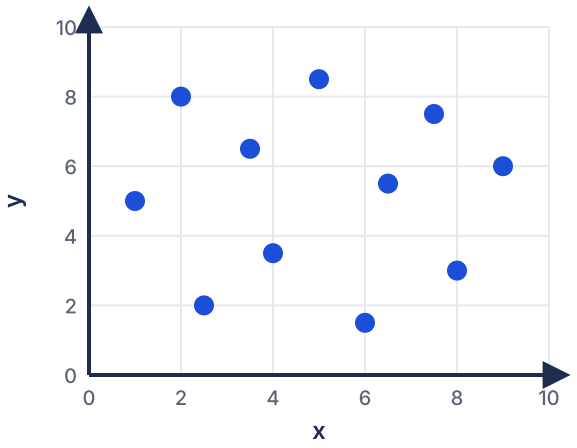
What kind of correlation does this scatter plot show?



(A) **Positive correlation**
(B) Negative correlation
(C) No correlation

2 Score 3 NC.8.SP.1

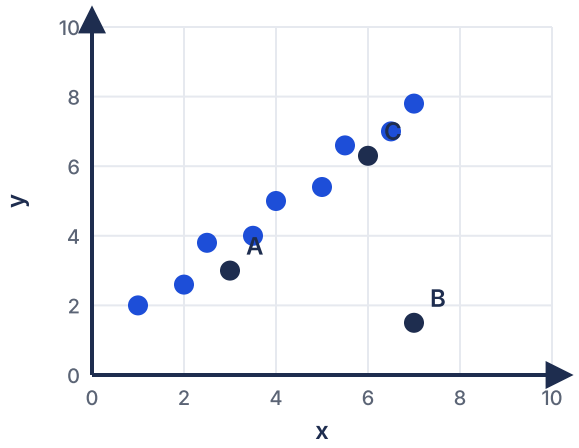
What kind of correlation does this scatter plot show?



(A) **No correlation**
(B) Positive correlation
(C) Negative correlation

3 Score 3 NC.8.SP.1

Which point is an **outlier**?



(A) **B**
(B) A
(C) C

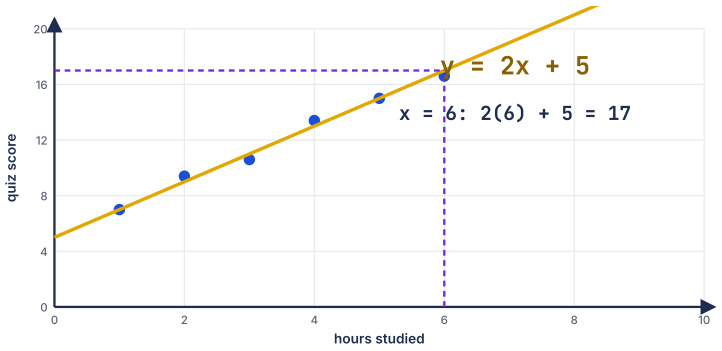
7.2 Line of best fit and predictions NC.8.SP.2

A **line of best fit** is a straight line through the middle of the dots. It is just **y = mx + b**: plug in an x to **predict** y.

Line of best fit
a line close to most dots, about as many above as below

Interpolation
predicting inside the data (safer)

Extrapolation
predicting outside the data (riskier)



Watch out: A good fit line does not have to touch any dot. It goes through the middle of the group.

Walk through:

Best fit: y = 3x + 4 (x = hours, y = score). Predict y for x = 5.

1. What do we do with x = 5?
   Put 5 in for x: y = 3(5) + 4 → y = 3(5) + 4
2. 3(5) = ?
   15 → y = 15 + 4
3. Finish.
   19 → y = 19

Check your understanding

1 Score 3 NC.8.SP.3

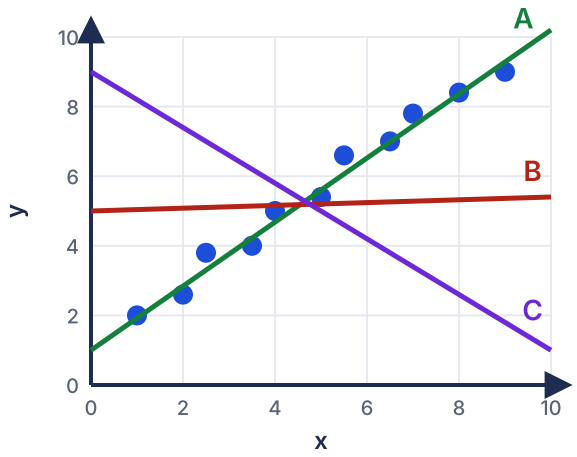
Best fit: $y = 2x + 5$. Predict y when x = 6.

Answer: 17

1. $y = 2(6) + 5$
2. $= 12 + 5 = 17$

2 Score 3 NC.8.SP.2

Which line fits the data best?



(A) Line A

(B) Line B

(C) Line C

7.3 Two-way tables NC.8.SP.4

A **two-way table** counts data sorted two ways. Inside boxes are **joint** counts; the edge totals are **marginal** counts. **Relative frequency** = count ÷ total, often as a percent.

Joint frequency

one inside box

Marginal frequency

a row or column total

Relative frequency

a count divided by a total: 12 ÷ 40 = 30%

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

	Likes dogs	Does not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

12 ÷ 20 = 0.60 = 60%

Watch out: Always ask "out of which total?" Out of everyone (40), or out of one row (20)?

Walk through:

Use the table. What percent of all students are girls who like dogs?

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

1. Find the box: girls who like dogs.
   15 → girls who like dogs: 15
2. Out of which total?
   All students: 40 → 15 ÷ 40
3. 15 ÷ 40 = ?
   0.375 = 37.5% → 15 ÷ 40 = 0.375 = 37.5%

Check your understanding

1 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

How many **boys** are there in all?

Answer: **20**

2 Score 3 NC.8.SP.4

	Likes dogs	Does not	Total
Boys	12	8	**20**
Girls	15	5	**20**
Total	**27**	**13**	**40**

What percent of **all** students do **not** like dogs?

Answer: **32.5%**

1. `does not like dogs: 13`
2. `13 ÷ 40 = 0.325 = 32.5%`